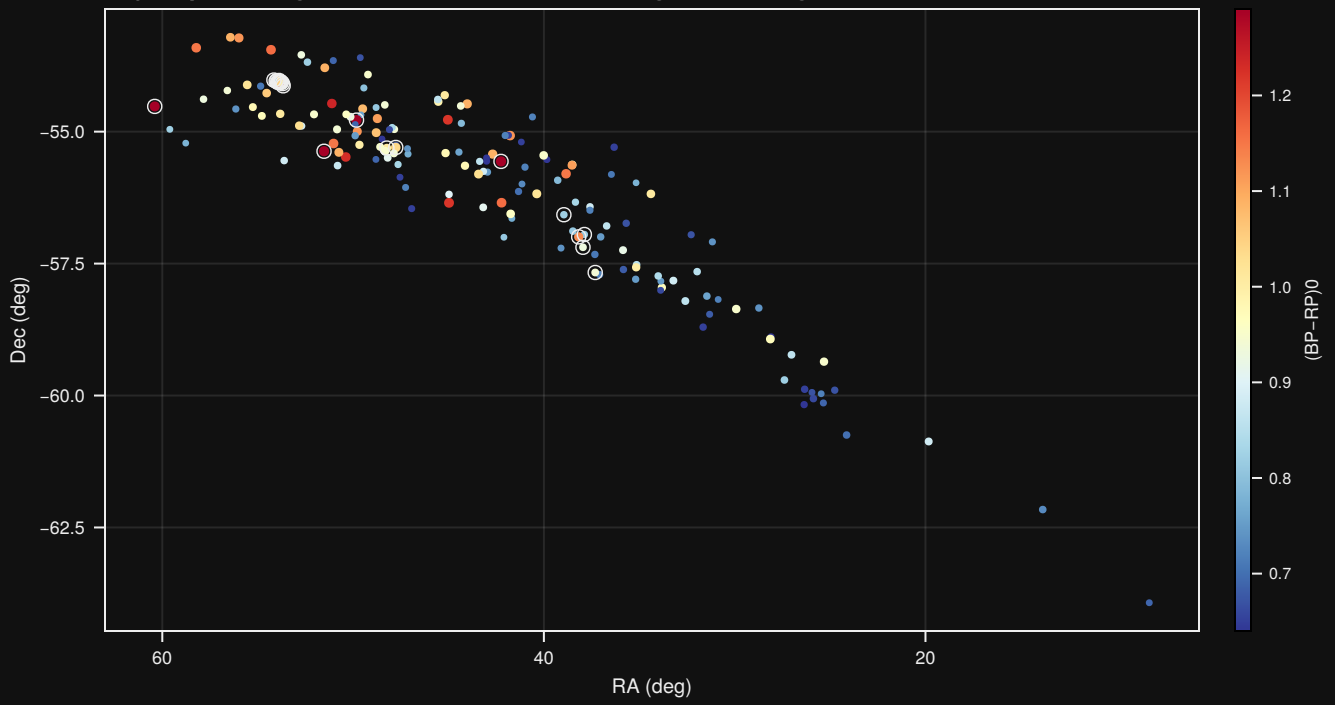
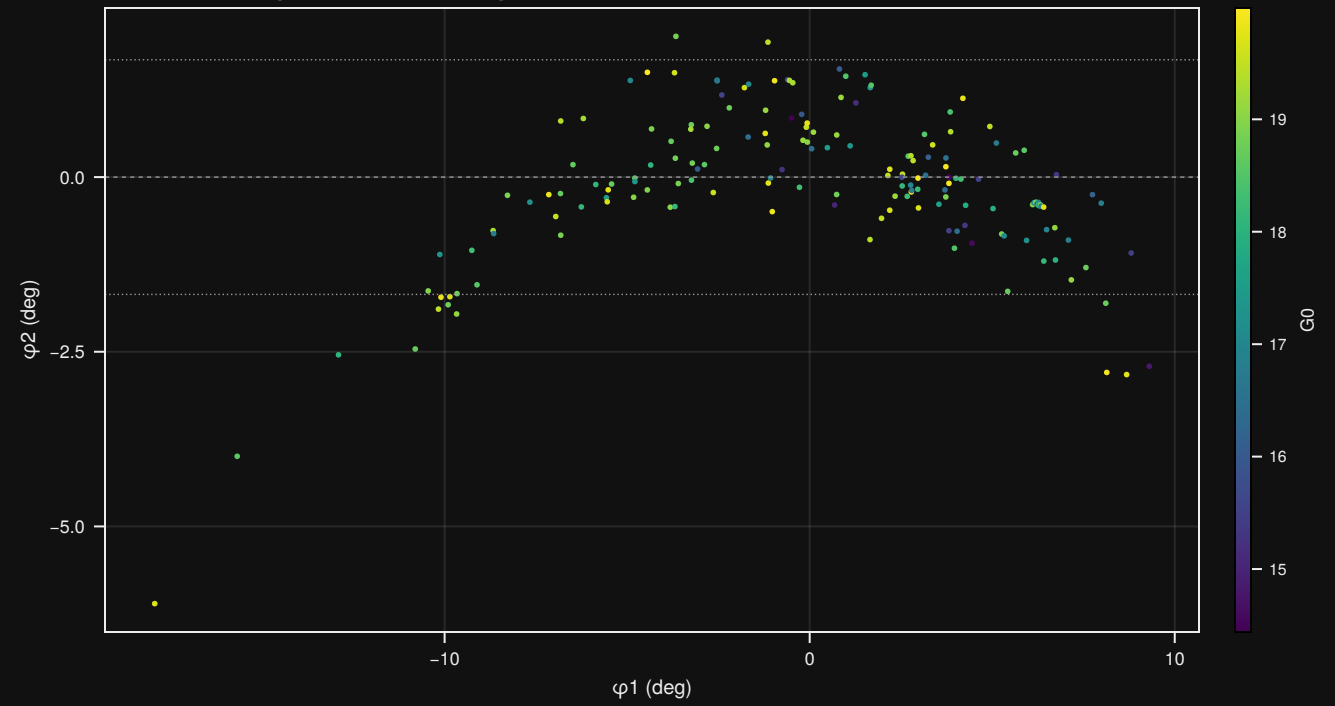
NGC1261

MEMBERSHIP	
Stars in catalog	173
with Gaia GSP-Phot	103 (60%)
with radial velocity	23
Stream id (sID)	12
GEOMETRY · great-circle fit	
Centroid RA, Dec	43.2, -55.5 deg
Centroid l, b	274.9, -53.8 deg
Pole RA, Dec	20.7, 31.7 deg
Length (φ1, 2–98%)	18.8 deg
Width σ(φ2), robust	0.84 deg
KINEMATICS	
pmRA* med ± σ	1.59 ± 0.34 mas/yr
pmDE med ± σ	-2.19 ± 1.01 mas/yr
μφ1* med ± σ	0.71 ± 0.26 mas/yr
μφ2 med ± σ	-2.75 ± 0.70 mas/yr
Parallax (cat.) med	0.041 ± 0.093 mas
Parallax (wtd, ZP-corr.)	0.059 ± 0.011 mas
Distance	d = 16.9 ± 3.2 kpc
Physical length	5.54 kpc
RV median ± σ(MAD)	63.5 ± 8.3 km/s
RV gradient dV/dφ1	3.61 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	14.4–20.0 (18.7)
(BP−RP)0 median	0.88
GSP-Phot [M/H] (n=103)	-1.52 ± 0.80
GSP-Phot Teff median	5253 K
RUWE > 1.4	0.0 %
PM clusters (largest)	9 (49 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	173 | 173
n_RV catalog | paper	23 | 23
<[Fe/H]> paper	-1.27
Stellar mass (paper)	8100 Msun
Pole offset vs paper	1.8 deg
Remark	Velocities messy, orbit apocen

Sky map · catalog members (n = 173) · size ∝ brightness · ring = has RV

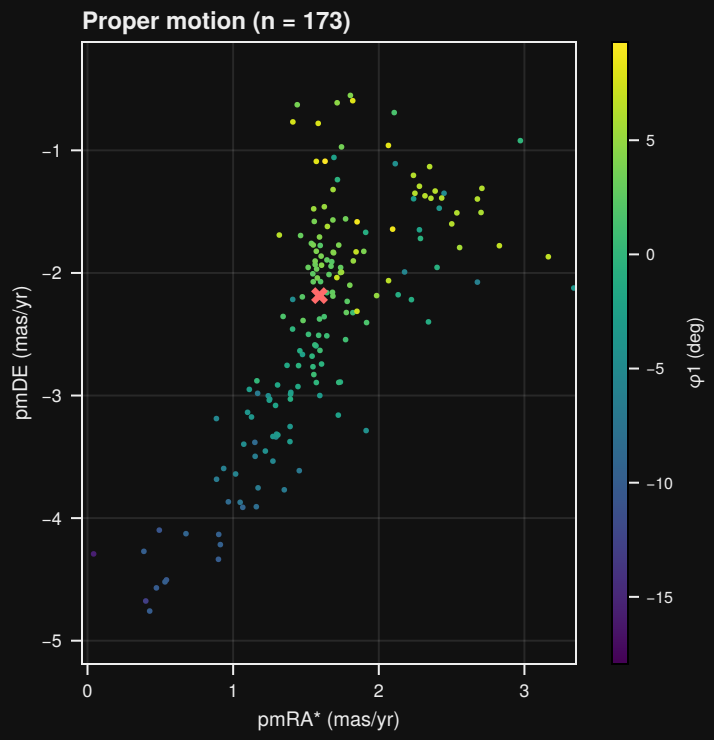


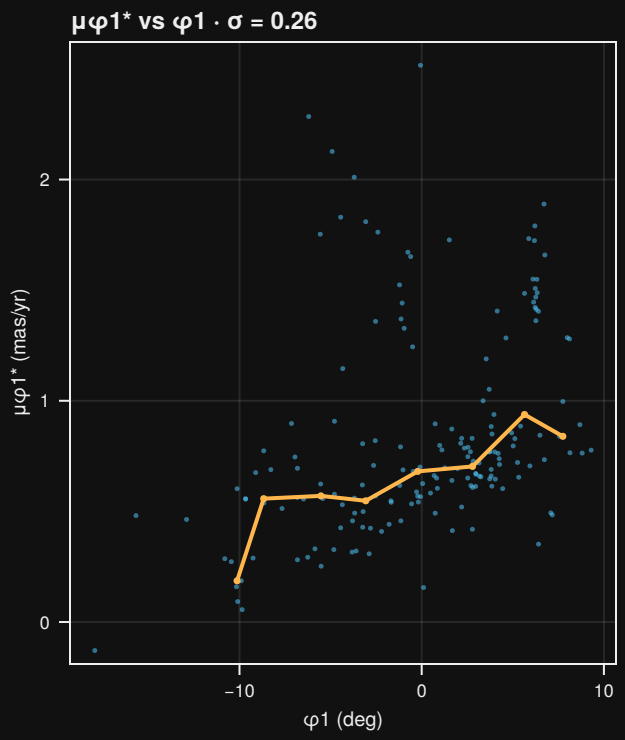
Stream frame (great-circle fit) · length 18.8° · width σ 0.84°

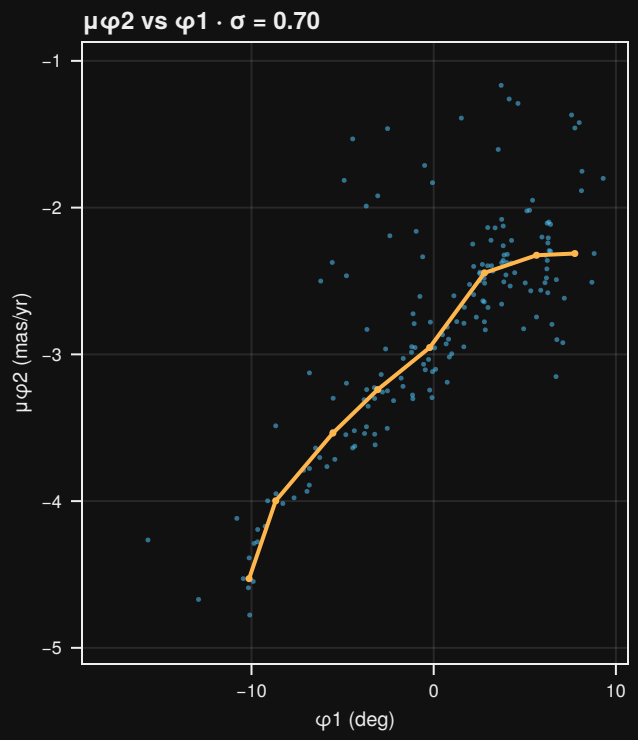


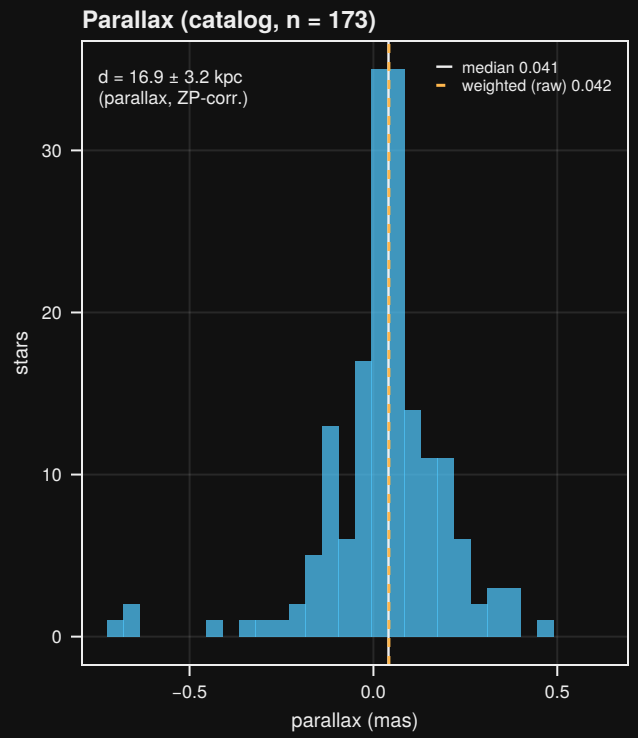
NGC1261

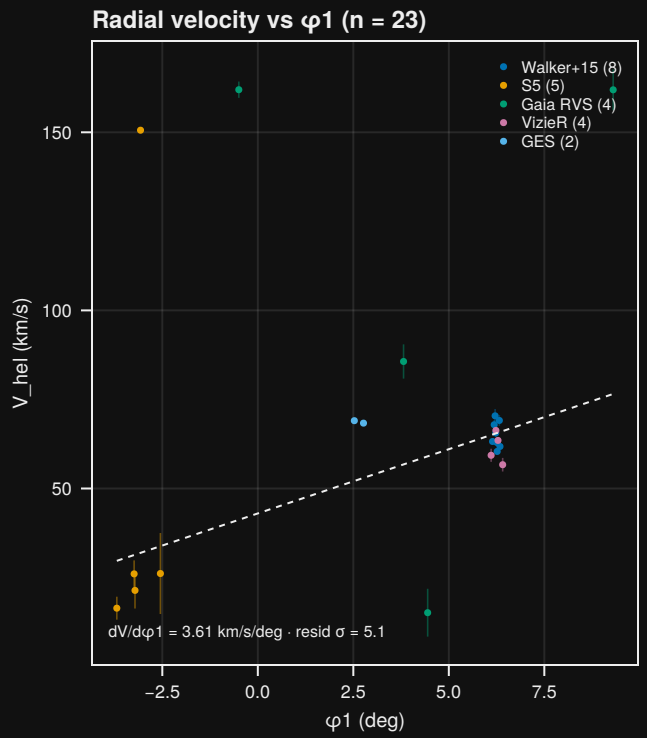
Kinematics · proper motion · parallax · radial velocity

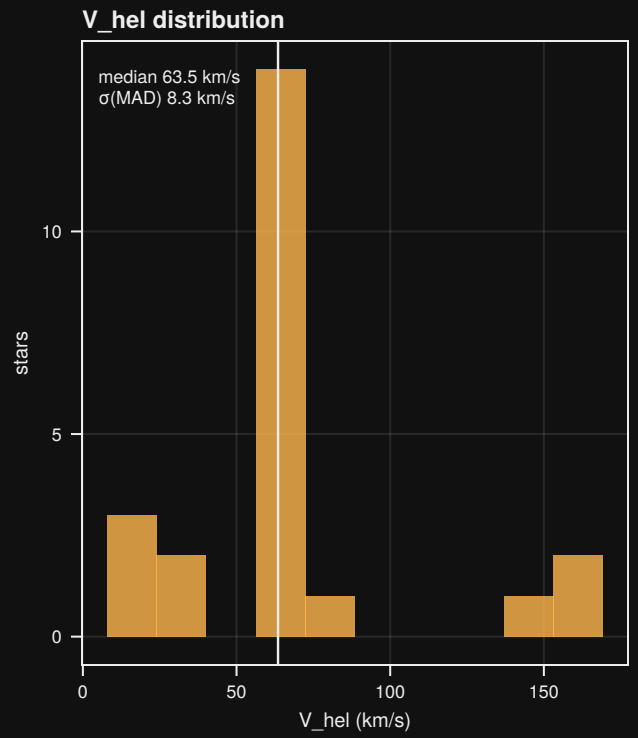






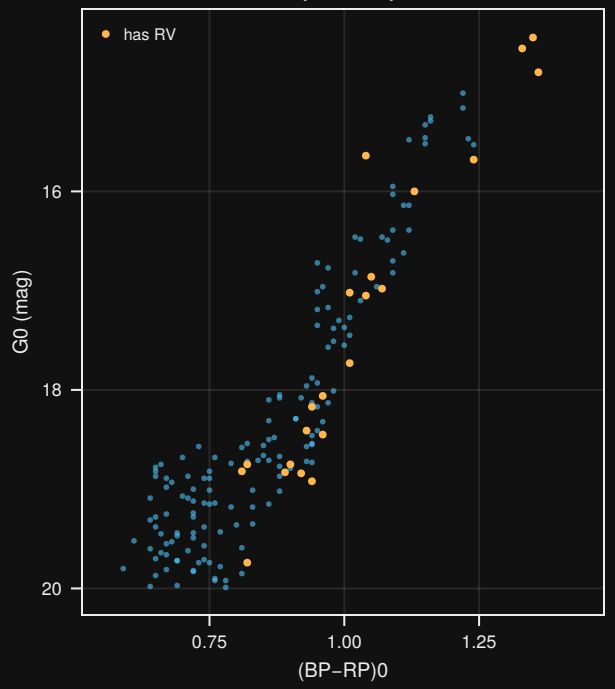




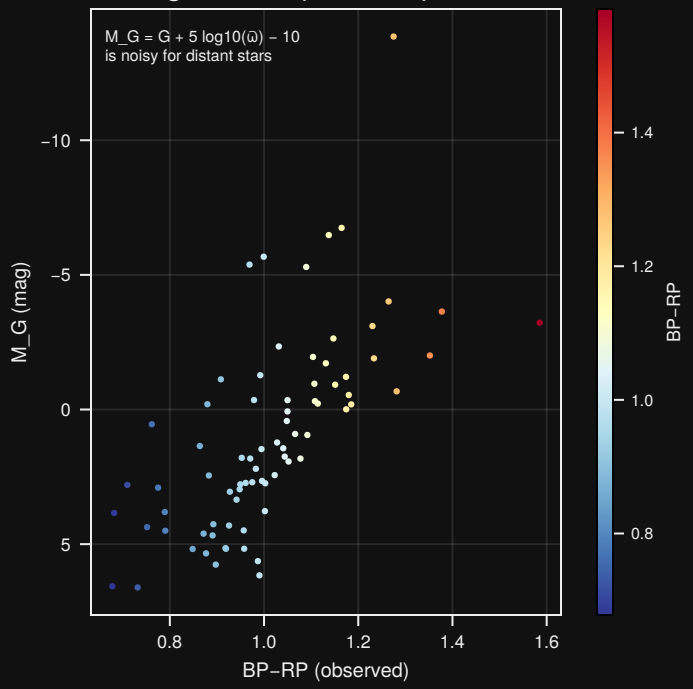


NGC1261

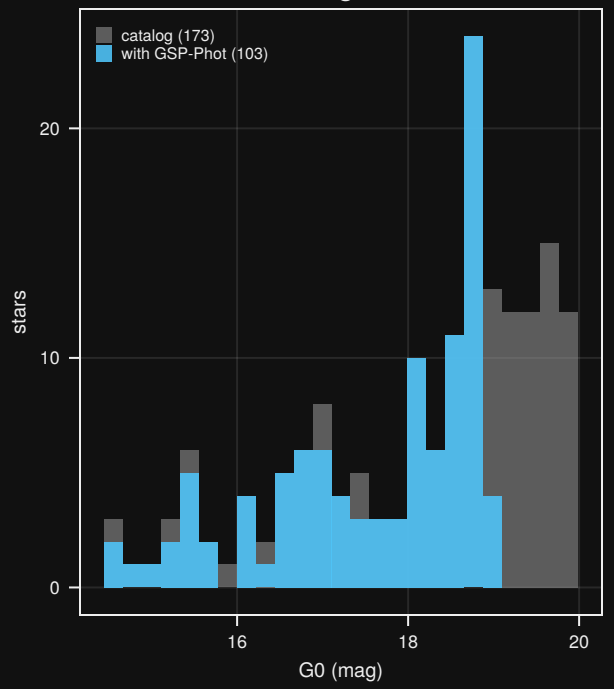
CMD · dereddened (n = 173)



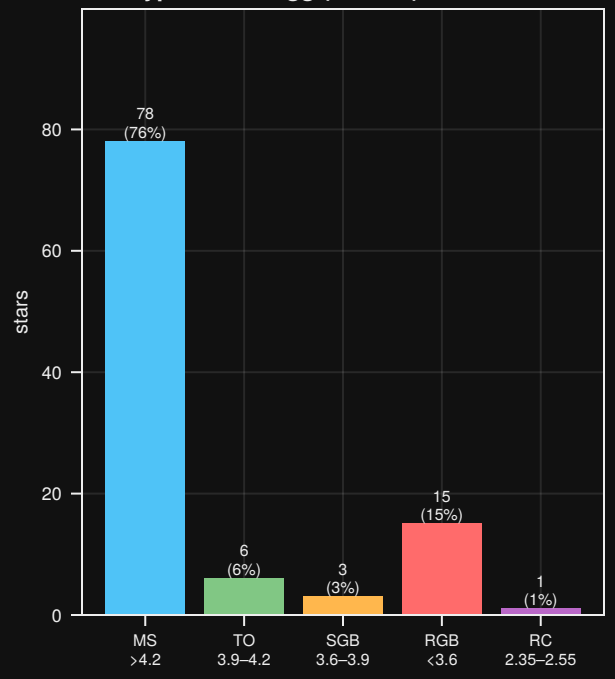
HR diagram · ϖ>0 (n = 77/103)



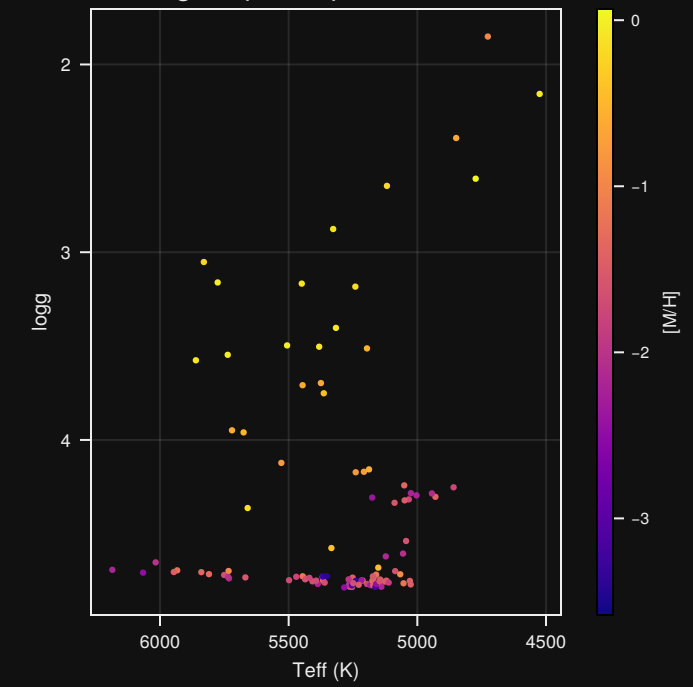
G0 & GSP-Phot coverage



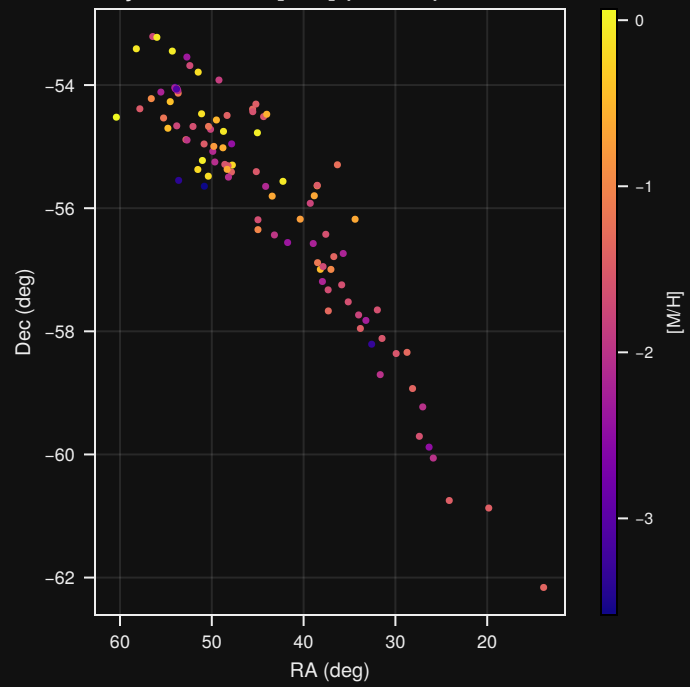
Star types from logg (n = 103)



Kiel diagram (n = 103)



Sky · GSP-Phot [M/H] (n = 103)

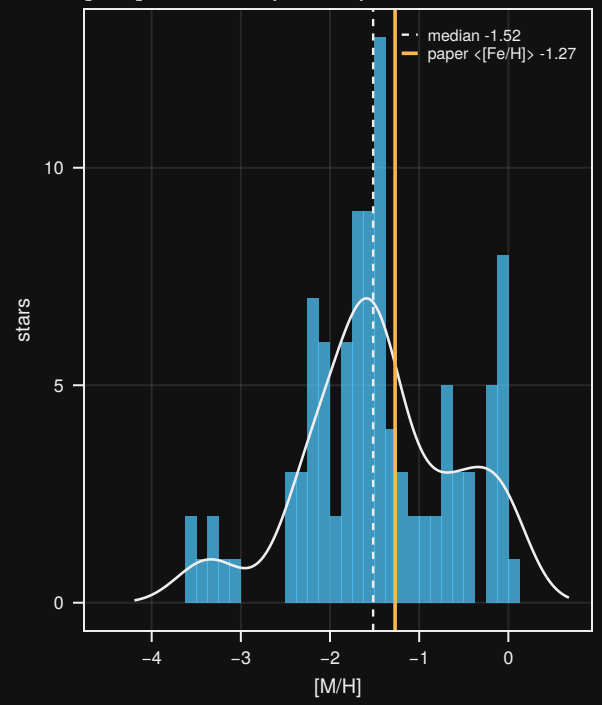


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

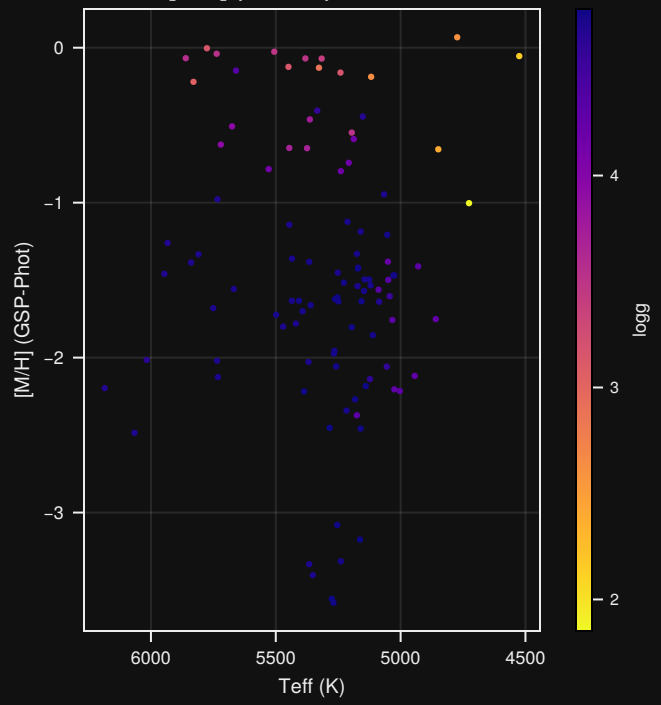
NGC1261

Metallicity · age · astrometric quality

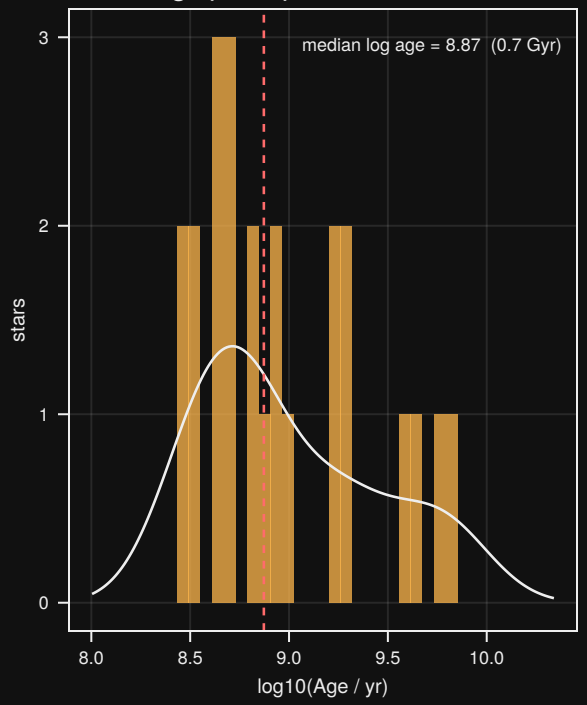
[M/H] GSP-Phot (n = 103)



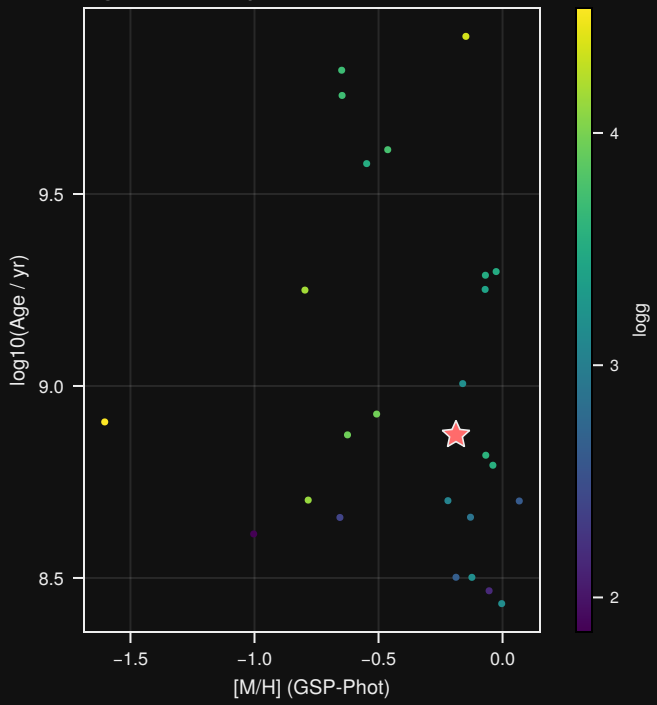
Teff vs [M/H] (n = 103)



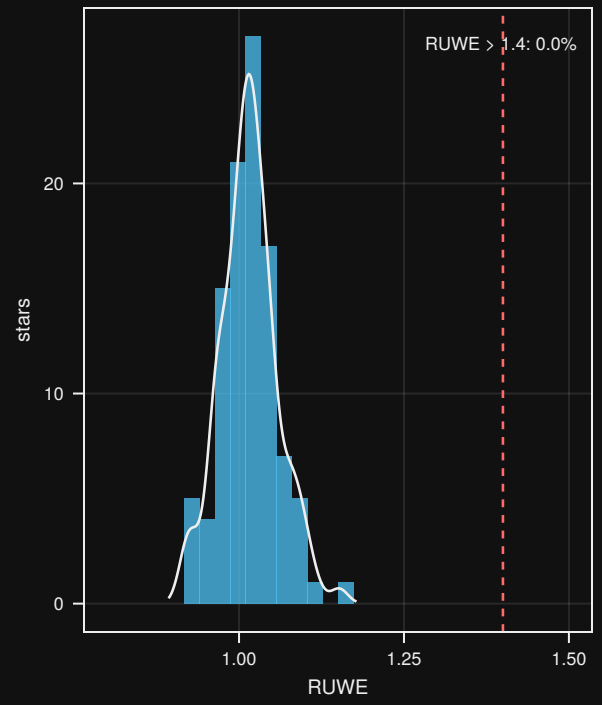
FLAME age (n = 25)



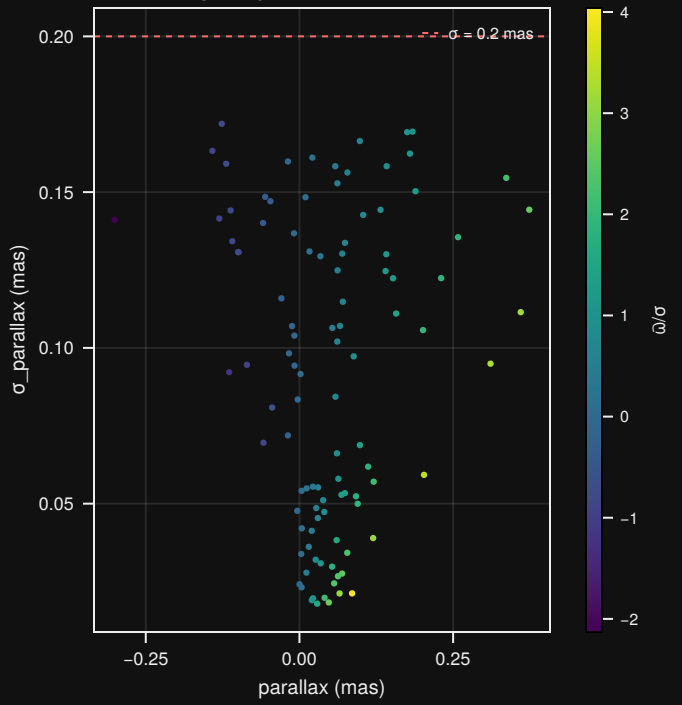
Age–metallicity (n = 25)



RUWE (n = 103)

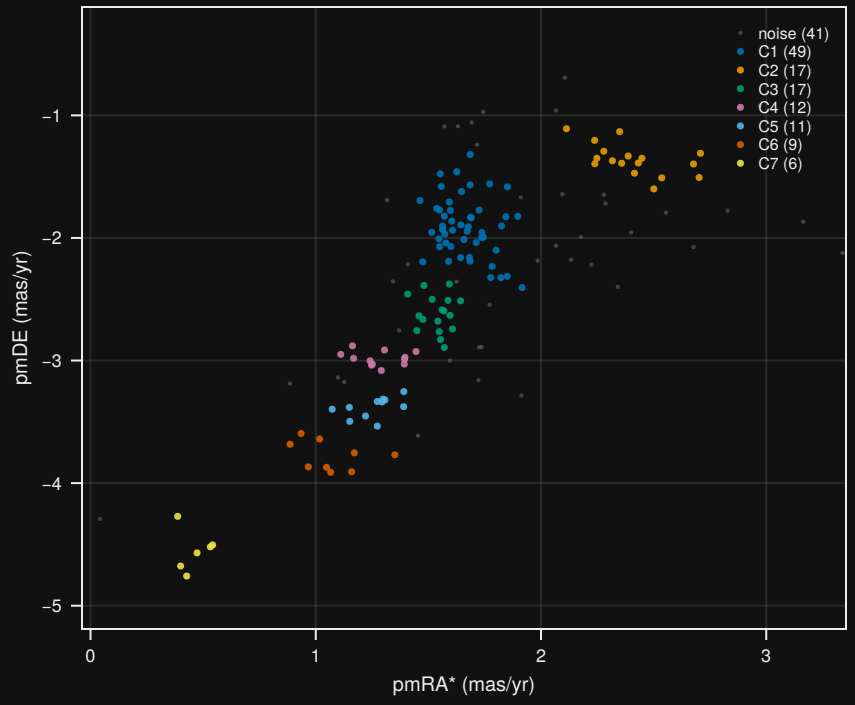


Parallax quality (n = 103)

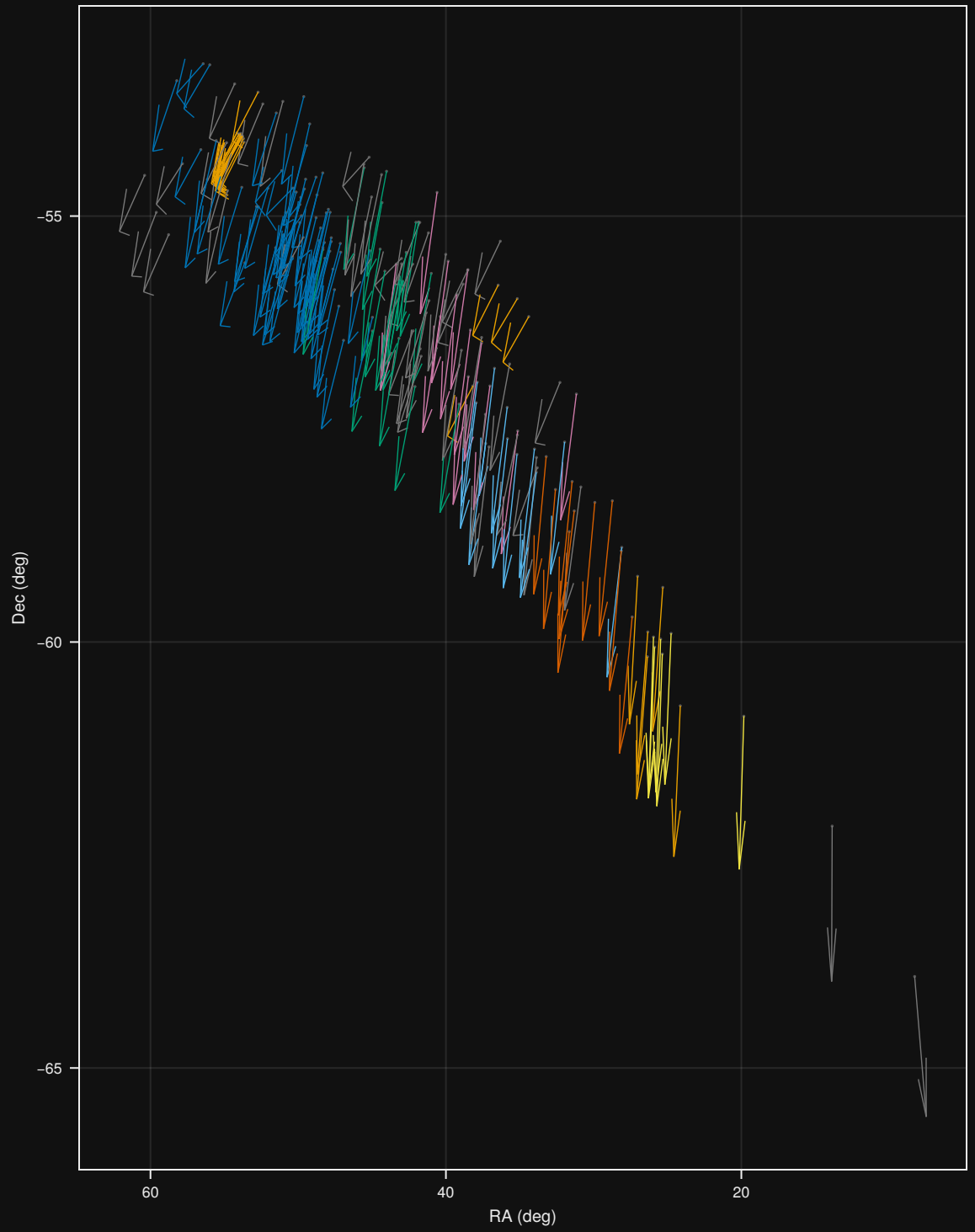


NGC1261

PM clustering · HDBSCAN · 9 cluster(s) · noise 24%



Proper-motion directions (arrows) coloured by cluster



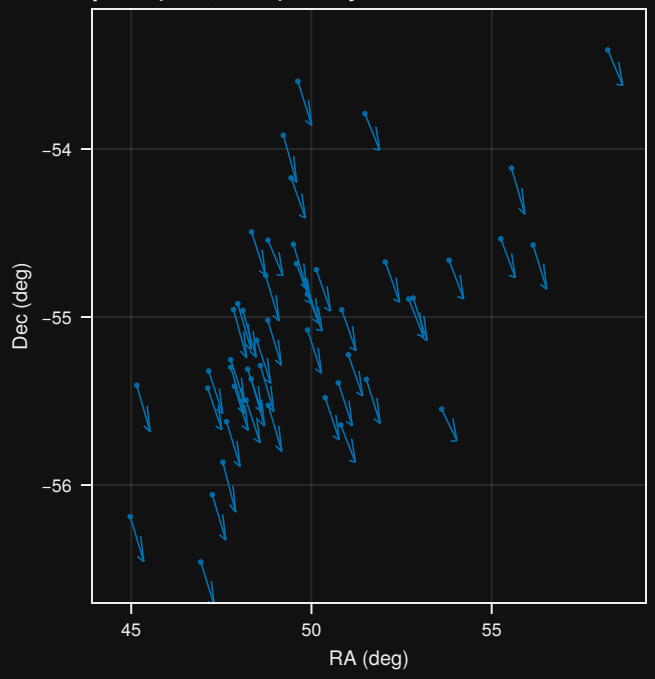
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	41	1.77	-2.12	0.54	-1.43	3
C1	49	1.66	-1.93	0.13	-1.50	4
C2	17	2.39	-1.37	0.17	-2.22	13
C3	17	1.55	-2.63	0.06	-1.30	0
C4	12	1.27	-2.99	0.16	-1.32	0
C5	11	1.27	-3.38	0.08	-1.64	3
C6	9	1.05	-3.77	0.16	-1.60	0
C7	6	0.45	-4.54	0.08	-1.73	0
C8	6	1.65	-0.62	0.24	-0.86	0

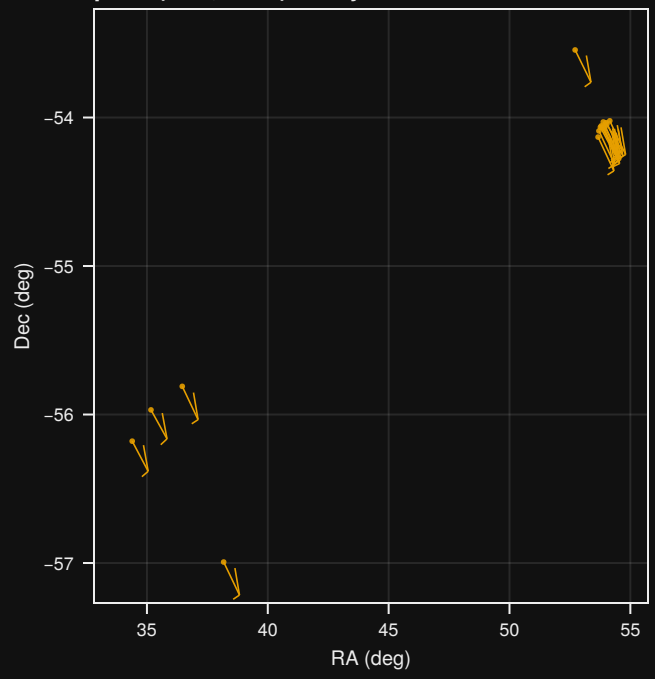
NGC1261

Per-cluster proper-motion vector maps

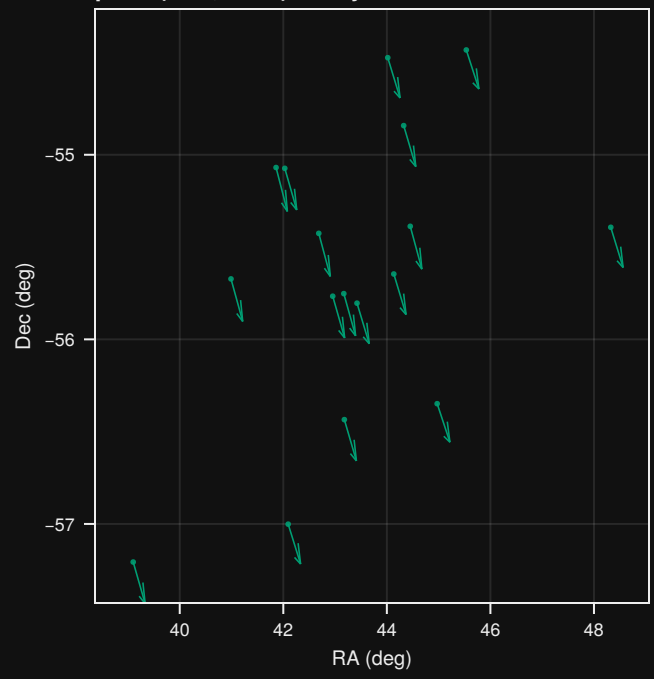
Cluster C1 · n = 49
pm = (1.66, -1.93) mas/yr



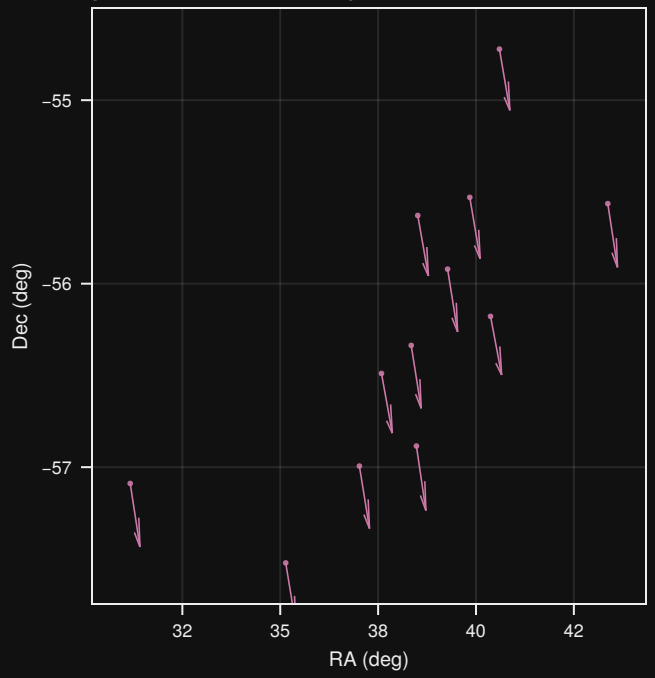
Cluster C2 · n = 17
pm = (2.39, -1.37) mas/yr



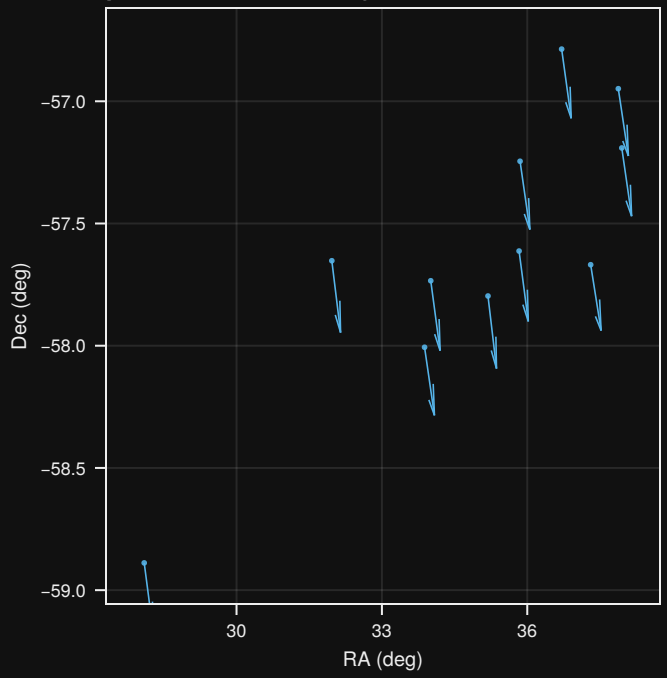
Cluster C3 · n = 17
pm = (1.55, -2.63) mas/yr



Cluster C4 · n = 12
pm = (1.27, -2.99) mas/yr



Cluster C5 · n = 11
pm = (1.27, -3.38) mas/yr



Noise · n = 41

